- **Technical consideration**

The clinical success of most amalgam restoration is highly dependent on the correct selection manipulation of the alloy and cavity design. If a restoration is defective, it is usually the fault of the operator and not the material.

Bonding of Amalgam: Amalgam does not adhere to the tooth structure. At best it affords only a reasonably close adaptation to the walls of the prepared cavity. For this reason cavity varnishes are used to reduce the gross leakage that occurs around a new filling.

The use of dentin bonding agents with amalgam is another relatively new method to reduce microleakage. If the amalgam is properly inserted, leakage decreases as the restoration ages in the mouth. This may be caused by corrosion products that form in the interface between the tooth and the filling, sealing the interface and preventing leakage.

Manipulation of Amalgam.

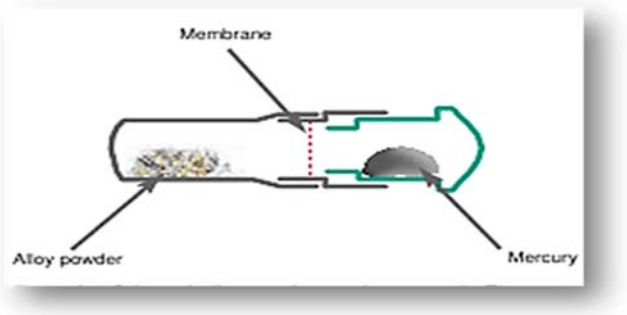
1. Proportion and Dispensers.
2. Trituration.
3. Condensation.
4. Carving.
5. Polishing.

1. Proportion and Dispensers: Because proportioning is important, manufacturers have developed some simple dispensers for alloy and mercury. Dispensers by volume are unreliable because it is affected by particle size and the degree of packing (trapped air and voids) in dispenser.

Tablets: This is most accurate method of dispensers manufacturers compress alloy powder into tablets of controlled weight which is used with measured amount of mercury.

Pre-proportioned capsules (disposable amalgam capsules) containing alloy particles and mercury in compartments separated by a membrane. Before use, the membrane is ruptured by compressing the capsule, and the capsule is then placed in a mechanical amalgamator.

Proportions of alloy to mercury: Some alloy require Hg-alloy ratio in excess of 1:1, whereas other use ratio of less than 1:1. the percentage of Hg varies from 43% to 54%.



2. Triturating (Mixing of Amalgam): Triturating of amalgam alloy and mercury is done with:

1. **Hand mixing** by use mortar and pestle.
2. **Mechanical mixing:** device called an amalgamator. Spherical or irregular low-copper alloy may be triturated at low speed (low energy), but most high copper alloys require high speed (high energy) .

Advantages of mechanical triturating:

1. Shorter mixing time.
2. More standardized procedure.
3. Requires less mercury when compared to hand mixing technique.

Mixing time: There is no exact recommendation for mixing time, since amalgamators differ in speed, oscillating pattern, and capsule designs. Spherical alloys usually require less amalgamation time than do lath-cut alloys. A large mix required slightly longer mixing time than a smaller one.

When amalgams with longer or shorter working times are desired, one should use amalgam alloys that are designed to react faster or slower and not attempt to achieve the change by altering the trituration time

Mixing Variables: Under mixing, normal mixing, or over mixing can result from variations in the condition of trituration of the alloy and mercury. The three types have different mechanical properties, dimensional change, strength and creep.

- **Under mixed amalgam** appears dull and is crumbly. A grainy under triturated mixture; restoration made of such a mixture has low strength and poor resistance to corrosion. The mixture may appear in solid mass, but the surface remains without luster.
- **Normal mix** appears shiny and separates in a single mass from the capsule. It appears rounded with a smooth shiny surface.
- **Over mixed amalgam** appears soupy and tends to stick to the inside of the capsule. It is shinier than that of the properly triturated one the mass appears flattened by the force of trituration.

Under **Normal** **Over**

3. Condensation The amalgam is placed in the cavity after triturating, and packed (condensed) using suitable instrument. It is adaptation of the amalgam mass to the cavity walls that controls the amount of mercury that will remain in the finished restoration. More mercury left in the mass after condensation, the weaker the alloy. Aims of condensation:

a. To adapt it to the cavity wall.
b. Remove excess mercury.
c. Reduce voids.

Irregularly shaped alloys needs higher percentage of mercury while with spherical alloys the amount of mercury is lower.

Increasing the condensation pressure results in a significant increase in compressive strength.

Condensation could be: Manual condensation or Mechanical condensation.

Manual Condensation: Circular condenser tips may prove adequate condensation. With irregularly shaped alloys small tip condenser (1 to 2) mm is use with high condensation forces in a vertical direction. While for spherical alloys one should use condensers with larger tips, almost as large as the cavity permits (particles tend to roll over one another).

Mechanical condensation: These devices are useful for condensing irregularly shaped alloys when high condensation forces are required.

Ultrasonic condensers are not recommended because during condensation they increase the mercury vapor level.

4. Carving: The filling is carved to reproduce the tooth anatomy. The carving should not be started until the amalgam is hard enough to offer resistance to the cavity instrument.

Burnishing: After the carving, the restoration is smoothened, by burnishing the surface and margins of the restoration. Burnishing is done with a ball burnished using light stroke proceeding from the amalgam surface to the tooth surface.

5. Polishing: Polishing minimize corrosion and prevents adherence of plaque. the polishing should be delayed for at least 24hours after condensation. For polishing wet abrasive powder in a paste form is used.

Mercury toxicity: Mercury is toxic, Free mercury should not be sprayed or exposed to the atmosphere. This hazard can arise during triturating, condensation, and finishing of restoration, and also during the removal of old restoration at high speed. Mercury vapors can be inhaled; skin contact with mercury should be avoided as it can be absorbed. It also reacts with gold.

Mercury has a cumulative toxic effect. Dentists and dental assistant are at high risk. Though it can be absorbed by skin or by ingestion, the primary risk is from inhalation. The clinic should be well ventilated. All excess mercury and amalgam waste should be stored in well-sealed containers.

An important aspect from a medical point of view, there is no clinical evidence to suggest that amalgam is causing illness in the general population.